REVISION HISTORY			
Rev	**Description of Change**	**Author**	**Submit Date**
1	Initial Release	Nicolas Greinert	09-28-2021

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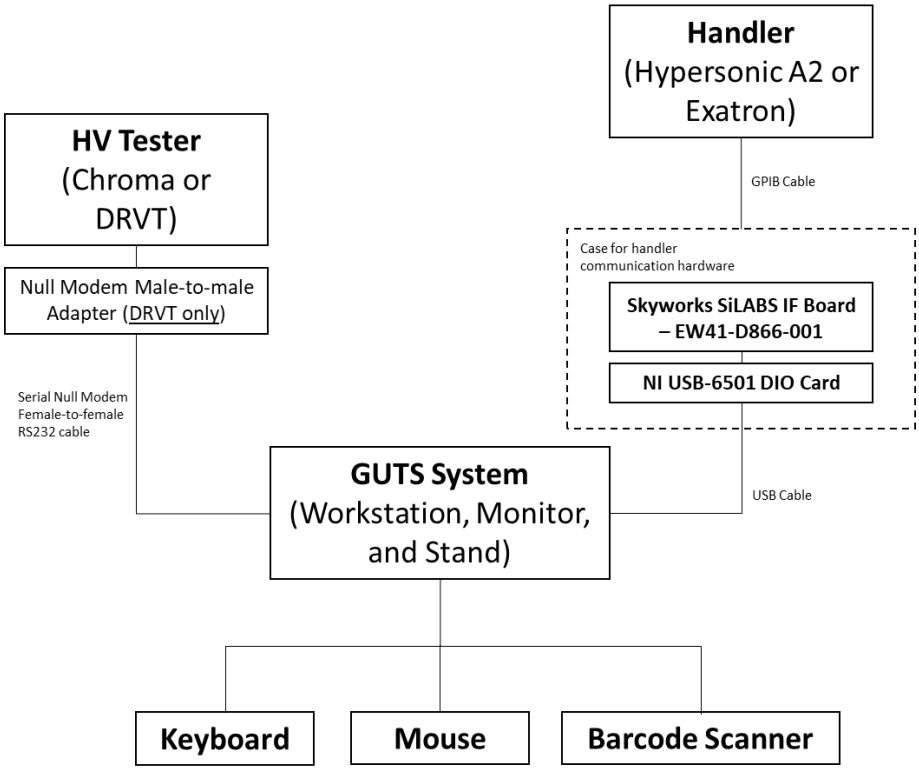
2.0 SYSTEM OVERVIEW – PURPOSE

The GUTS Hi-pot system is the PC that will control the DRVT/CHROMA Hi-pot testers as-well as the HV handlers. This document outlines the operator guide to perform the HV test. This document is intended to be used by the operator. There is a separate document that it intended to be used by the system engineer/administrator/technician.

This document is intended to outline the initial hardware/software setup prior to testing as-well as how to execute the testing itself. This document assumes that the Hi-pot tester is already correctly connected to the handler and both the Hi-pot tester and the handler are powered on.

2.1 SYSTEM OVERVIEW – HARDWARE DIAGRAM



2.2 SYSTEM OVERVIEW – HARDWARE COMPONENTS

Below is all of the required hardware for the GUTS system:

- 1x Dell 3450 Small Form Factor Workstation
- 1x Dell 24-inch Monitor
- 1x Dell Small Form Factor All-in-one Stand
- 1x Case for handler communication hardware
    - The case contains the following:
        - 1x NI USB-6501 DIO Card
        - 1x Skyworks SiLABS IF Board – EW41-D866-001
- 1x Keyboard
- 1x Mouse
- 1x Barcode Scanner
- 1x GPIB cable
- 1x Serial Null Modem Female-to-female RS232 cable
- 1x Null Modem Male-to-male Adapter (For every DRVT)

3.0 HARDWARE SETUP – COMMUNICATION HARDWARE OPTIONS

In order to test with the GUTS HV setup, one Hi-pot Tester and one Handler must be connected to the GUTS PC. Below are the the currently supported Hi-pot Tester and Handler options and their communication types in parenthesis:

Hi-pot Tester:

- Chroma 19501-K (RS232)
- DRVT-1000 (RS232)

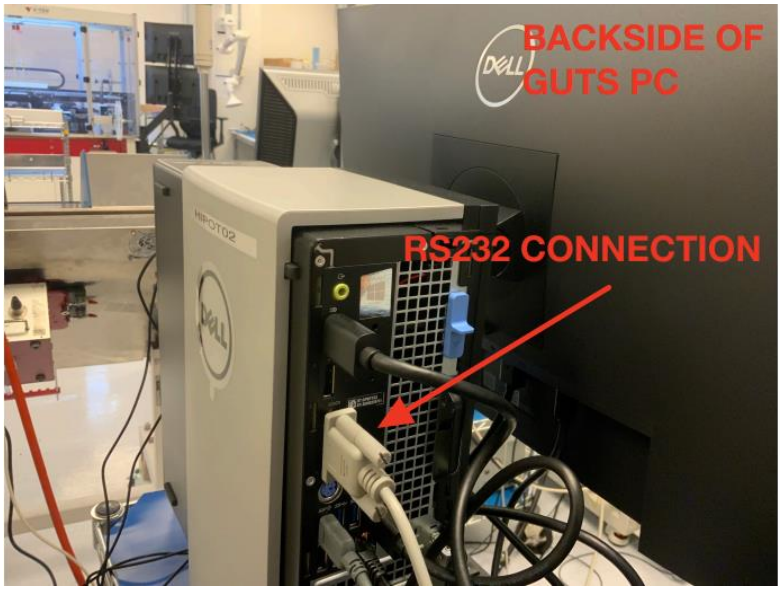
Handler:

- Exatron 3000 (TTL)
- Exatron 5000 (TTL)
- Exatron 8000 (TTL)
- Hypersonic A2 (TTL)
- Hontech (GPIB) – In Progress
- NS7000 (GPIB) – In Progress

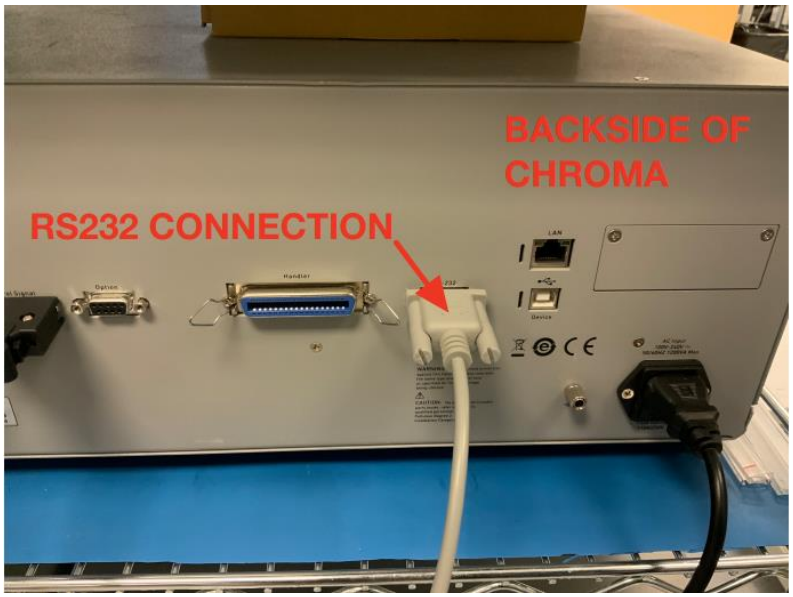
3.1 HARDWARE SETUP – HIPOT COMMUNICATION HARDWARE

The Hi-pot testers communicate via RS232. **A serial null modem female-to-female RS232 cable must be used to connect the GUTS PC (using the serial port on the back of the PC) to the Hi-pot tester.**

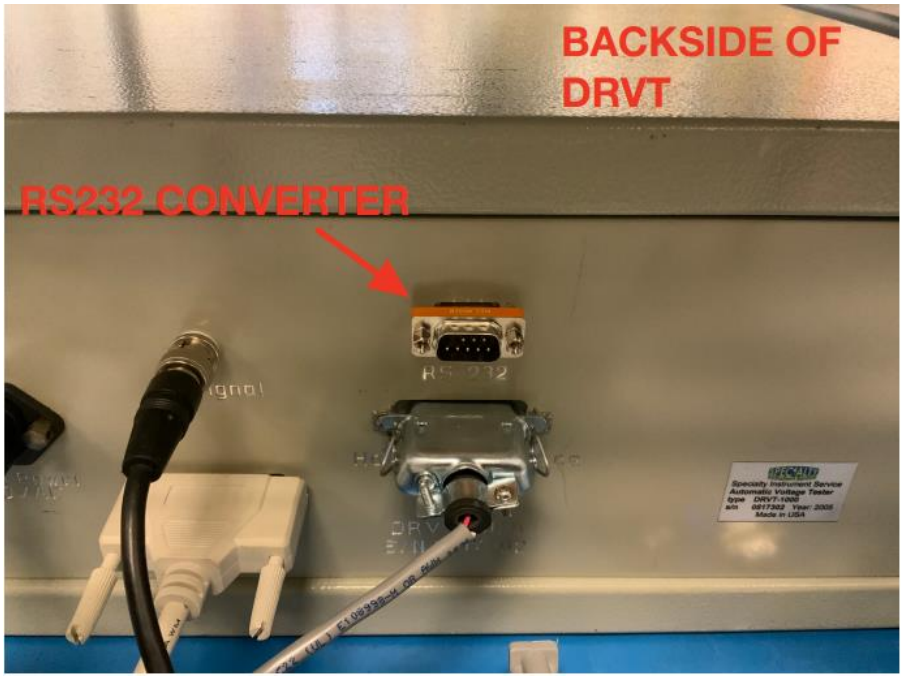
RS232 connection to GUTS PC:

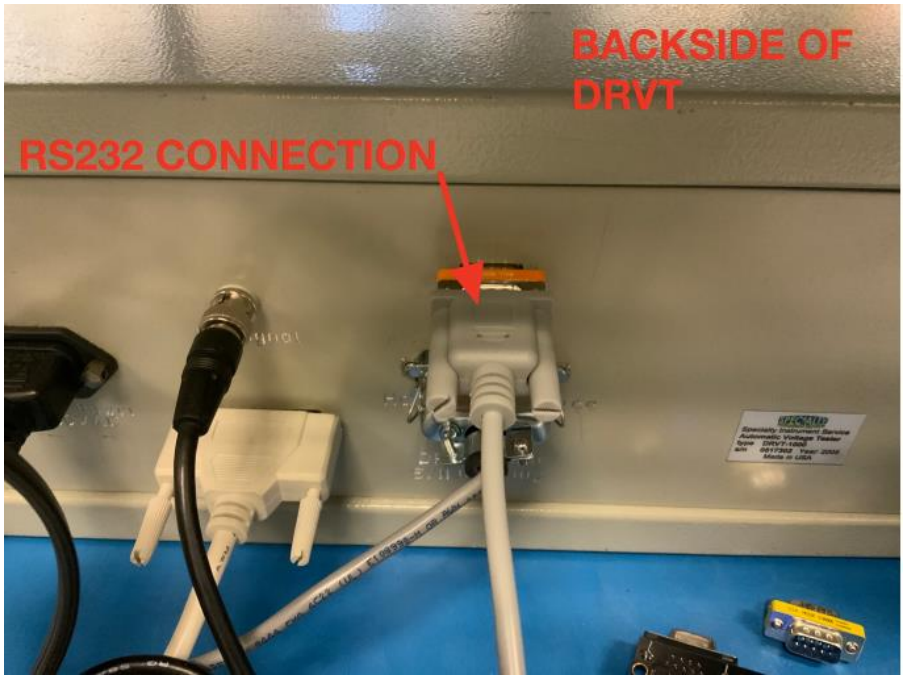


RS232 connection to Chroma 19501-K:



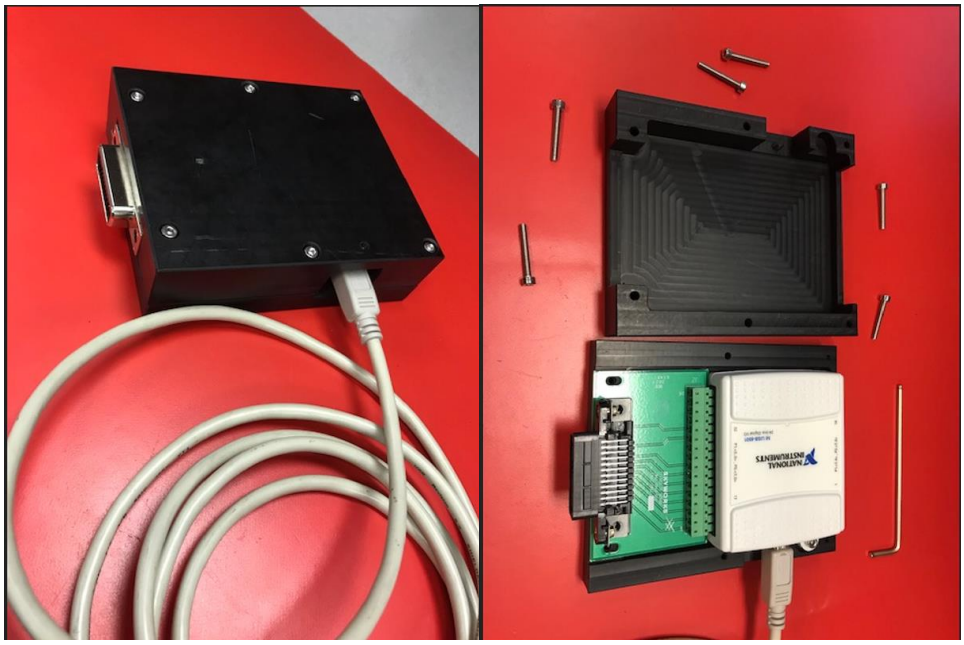
RS232 connection to DRVT-1000 (**NOTE: DRVT-1000 requires a null modem male-to-male adapter**):





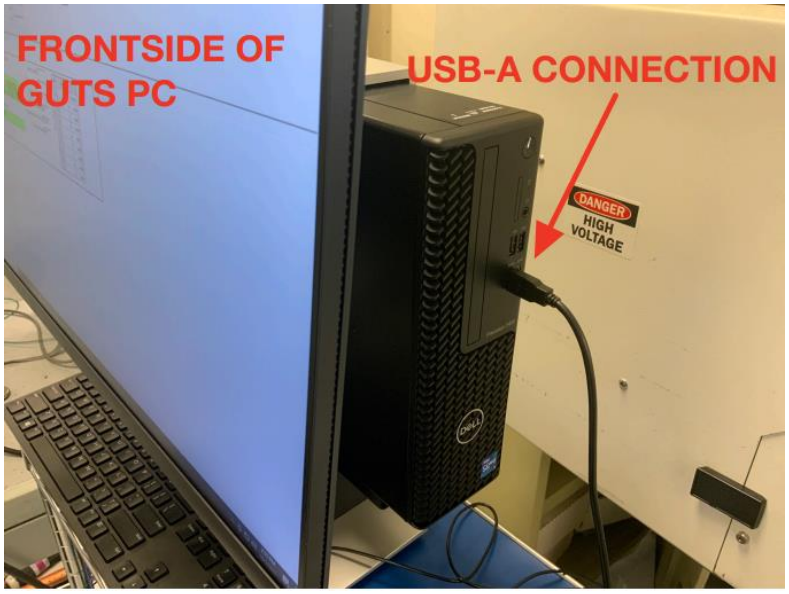
3.2 HARDWARE SETUP – HANDLER COMMUNICATION HARDWARE

The Exatron 3000, Exatron 5000, Exatron 8000, and Hypersonic A2 communicate via TTL. An NI USB-6501 DIO Card, Skyworks SiLABS IF Board – EW41-D866-001, GPIB cable, and USB-B <-> USB-A cable are required to connect the GUTS PC to the Handler. The NI USB-6501 DIO Card and Skyworks SiLABS IF Board – EW41-D866-001 are inside of a case:

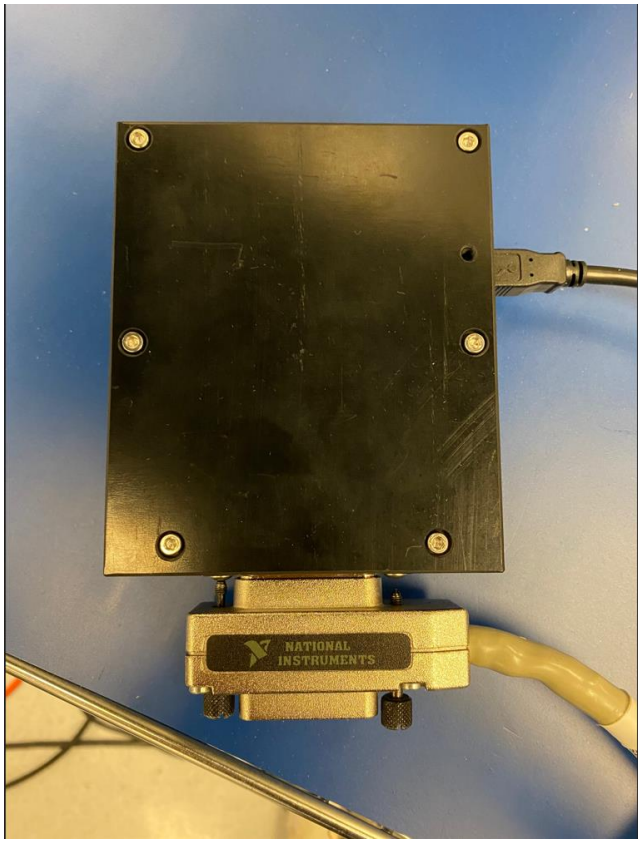


The connections to be made are as follows.

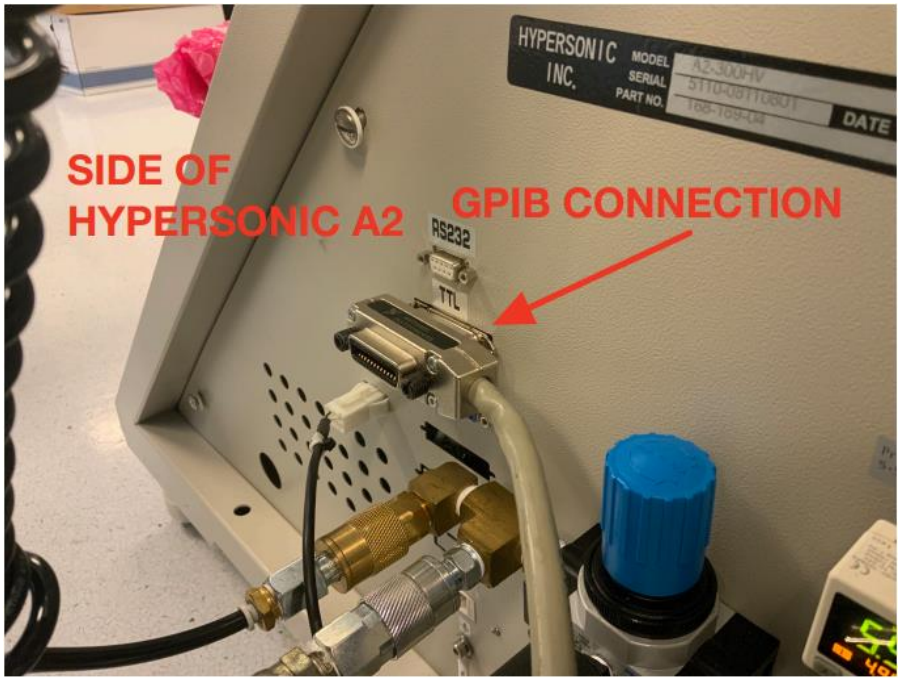
USB-A connection to GUTS PC:



USB-B and GPIB connection to handler hardware case:

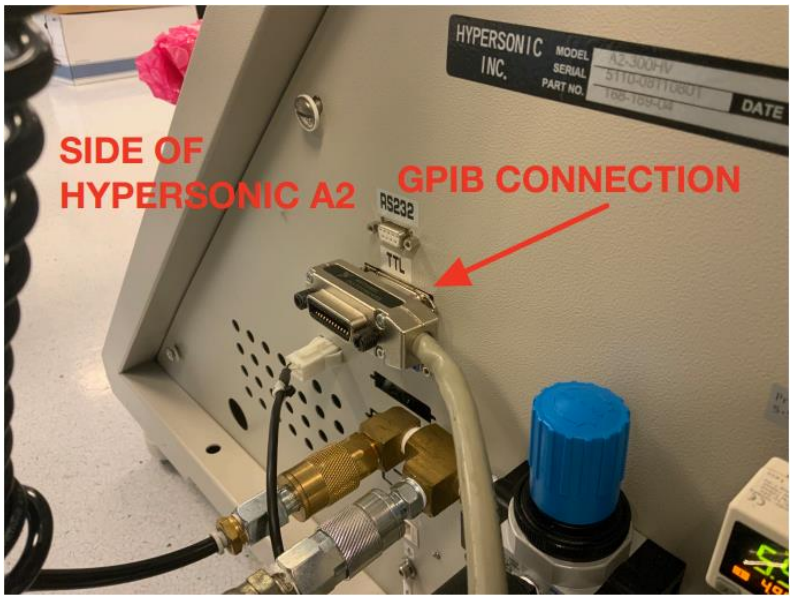


GPIB connection to handler:



The Hontech and NS7000 handlers communicate via GPIB. Only certain GUTS PC's can be used with the Hontech and NS7000 handlers. Please contact engineering if you are unsure of which GUTS PC's can support these handlers. The GUTS PC's that can support the Hontech and NS7000 handlers will have a GPIB port installed on the back that the GPIB cable will plug into.

GPIB connection to handler:

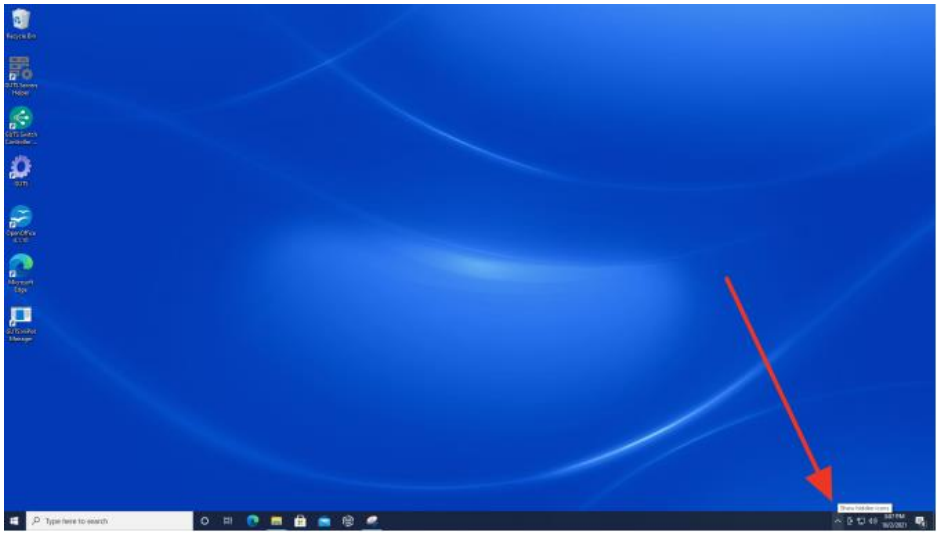


4.0 GUTS PC SETUP – LOGIN AND HANDLER DRIVER CHECK

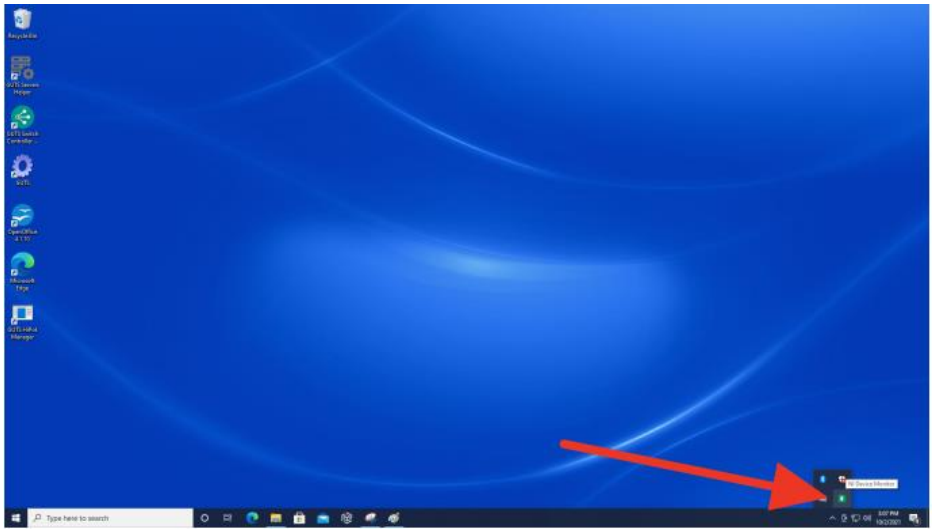
Once the Hi-pot tester and handler connections are complete, power on the GUTS PC and login with the account provided by your network administrator.

Once you are logged into the GUTS PC, if you are using one of the TTL handlers (Exatron 3000, Exatron 5000, Exatron 8000, or Hypersonic A2), the handler driver communication port should be checked, otherwise you may skip to section 5.0. This can be checked with the following steps:

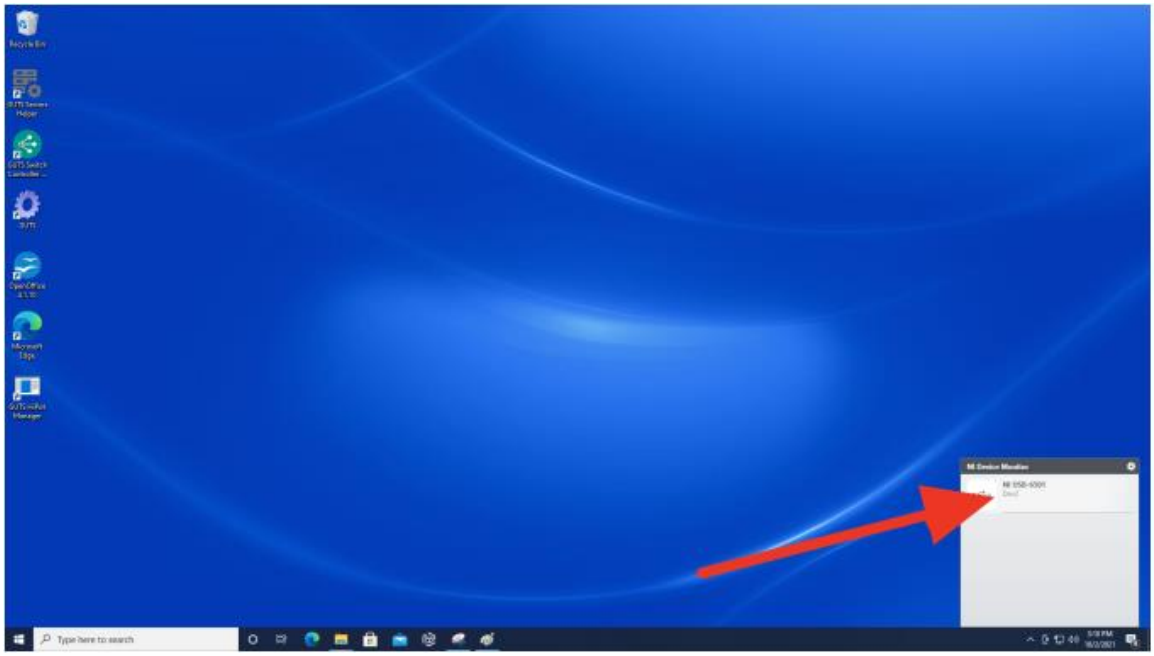
1. From the desktop, click “Show hidden icons” on the bottom right:



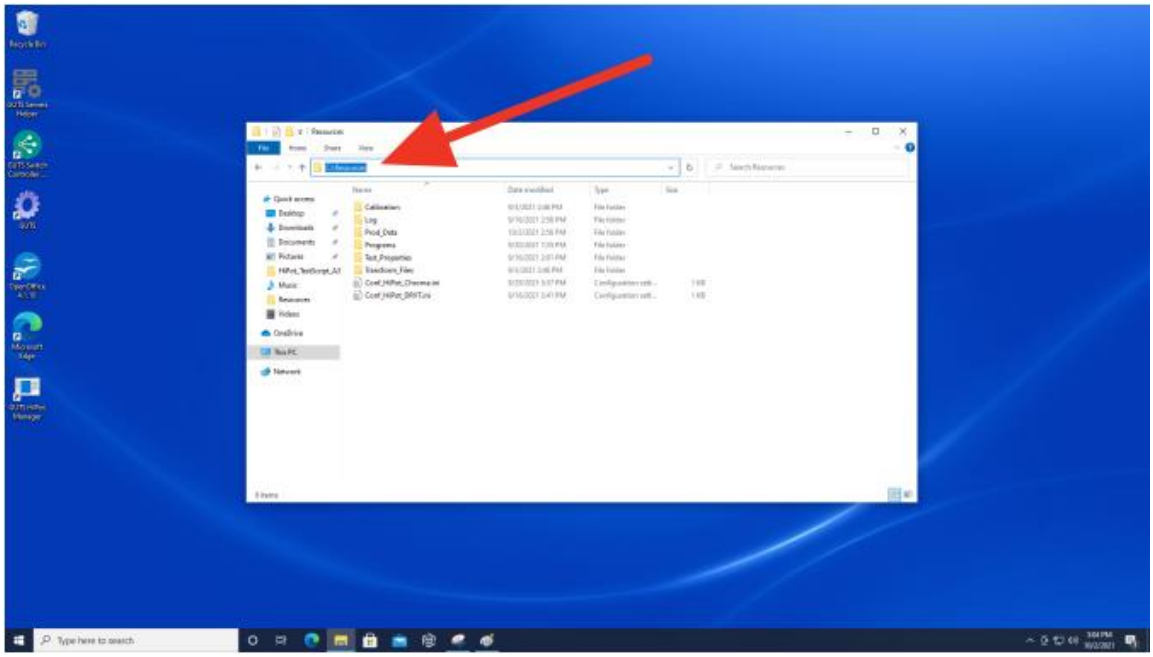
2. Click the “NI Device Monitor” icon:



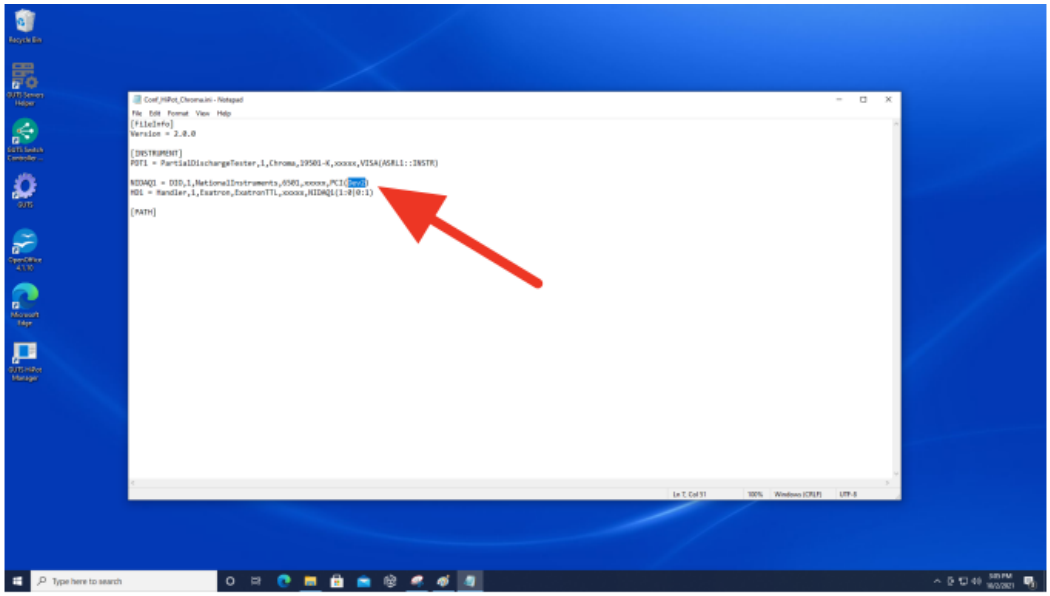
3. Note down the “DevX” number (i.e. “Dev2”):



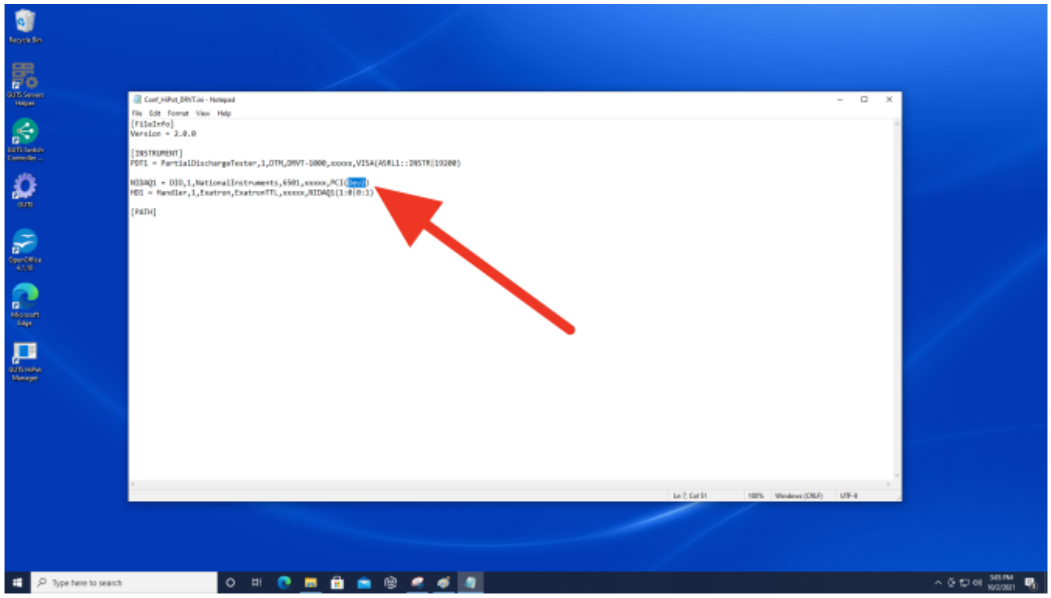
4. Navigate to C:\Resources:



5. Open the “Conf_HiPot_Chroma.ini” file in Notepad and make sure that the “DevX” number matches the one that was noted down earlier. If it does not match, change the value in this file to match the one displayed on the “NI Device Monitor”:



6. Save & Close the “Conf_HiPot_Chroma.ini” file
7. Open the “Conf_HiPot_DRVT.ini” file in Notepad and make sure that the “DevX” number matches the one that was noted down earlier. If it does not match, change the value in this file to match the one displayed on the “NI Device Monitor”:

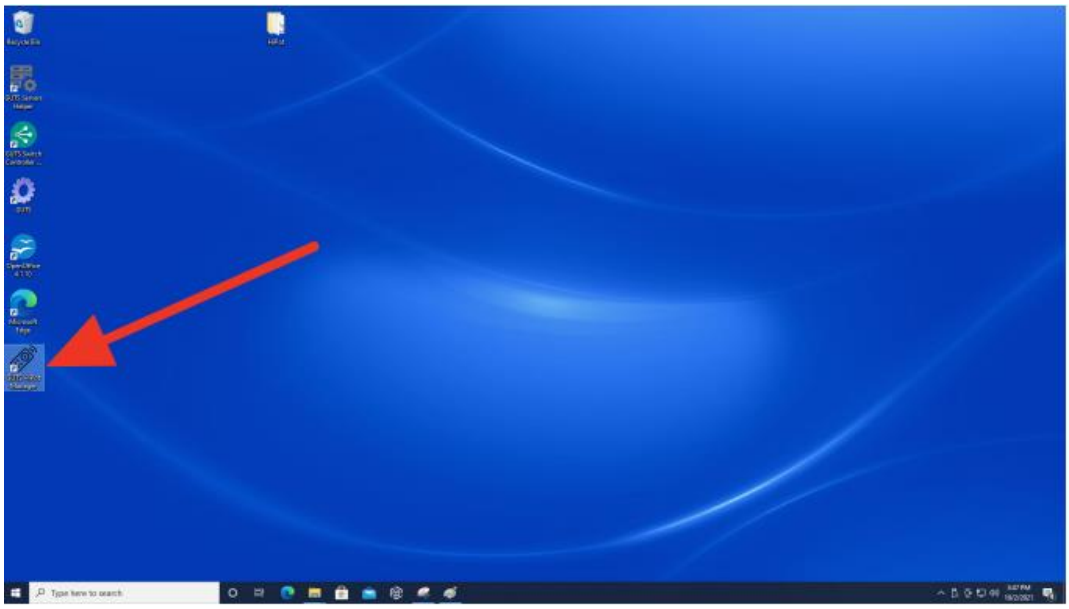


8. Save & Close the “Conf_HiPot_DRVT.ini” file

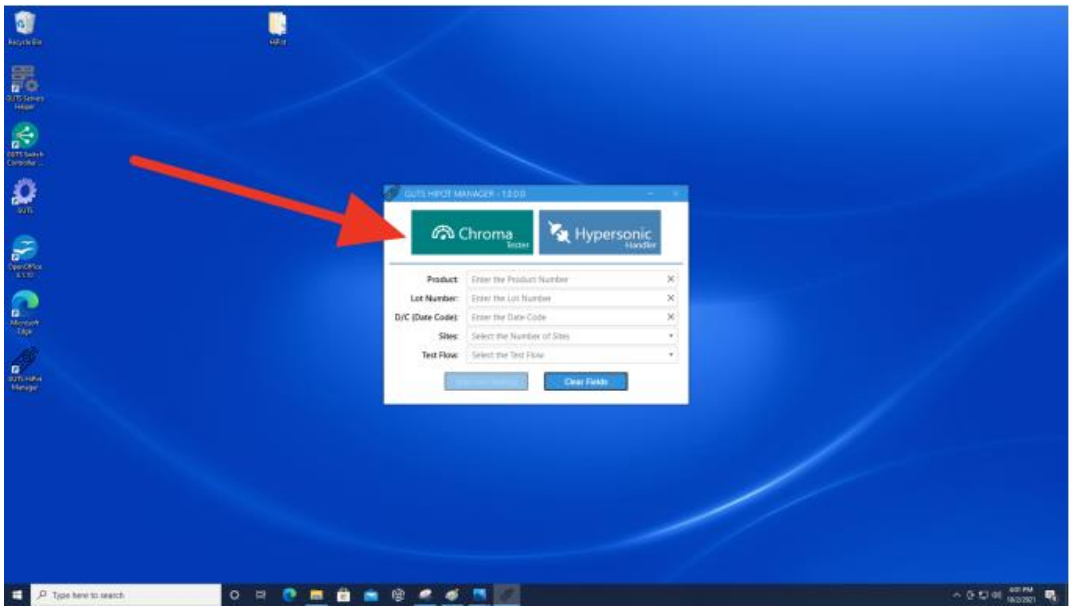
5.0 TESTING PARTS – STARTING A NEW LOT

At this point, the Hi-pot tester, handler, and GUTS PC should be ready for testing. The following steps outline how to start a new lot:

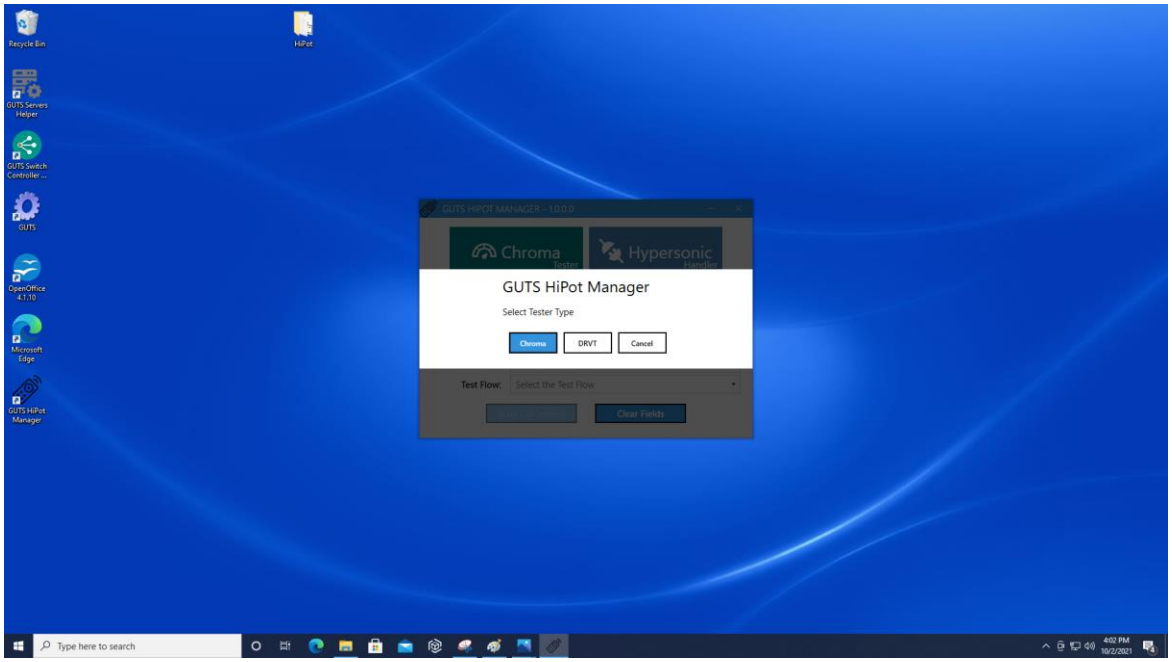
1. Open the “GUTS HiPot Manager” application:



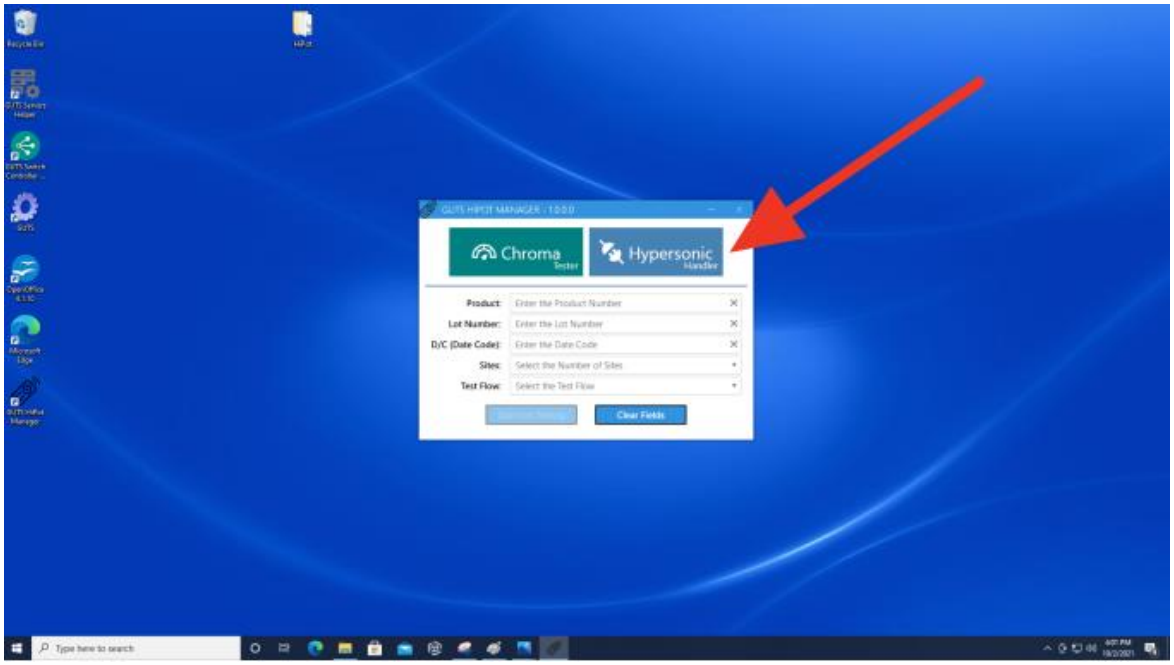
2. Select the green box on the top left:



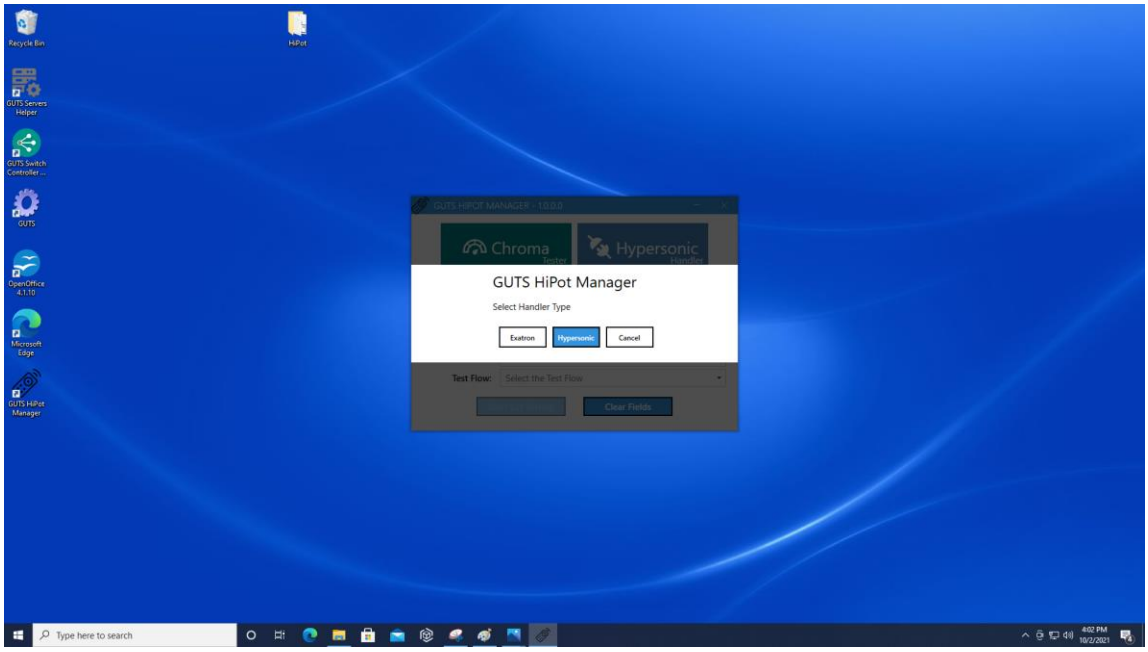
3. Select the Hi-pot tester that will be used for testing:



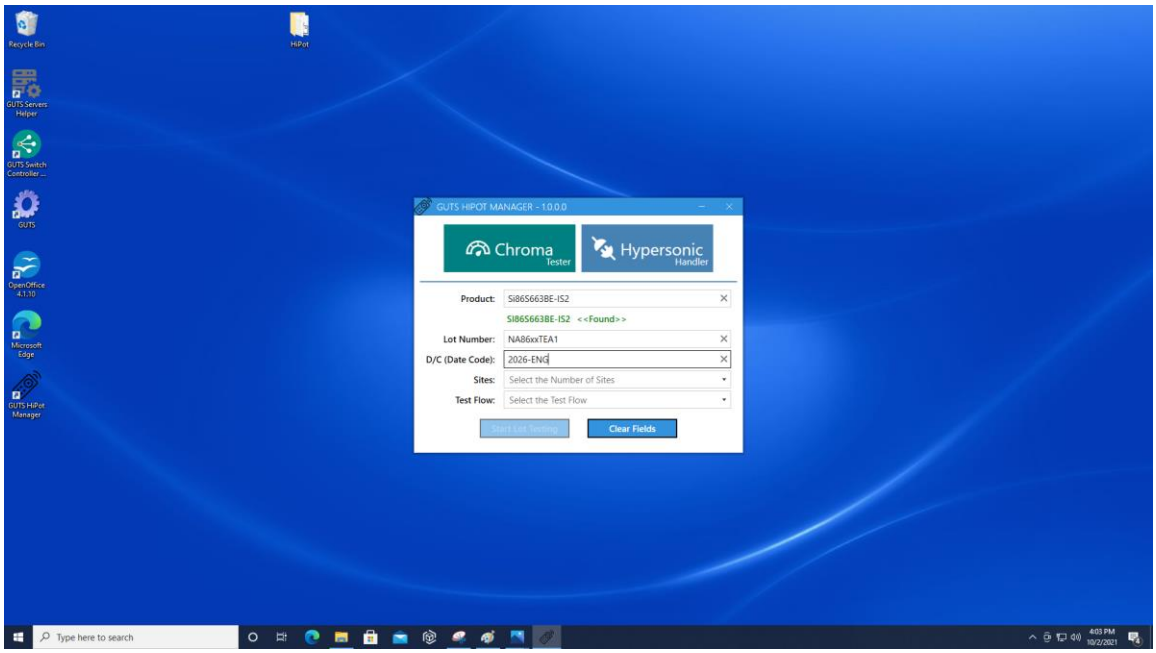
4. Select the blue box on the top right:



5. Select the Handler that will be used for testing:

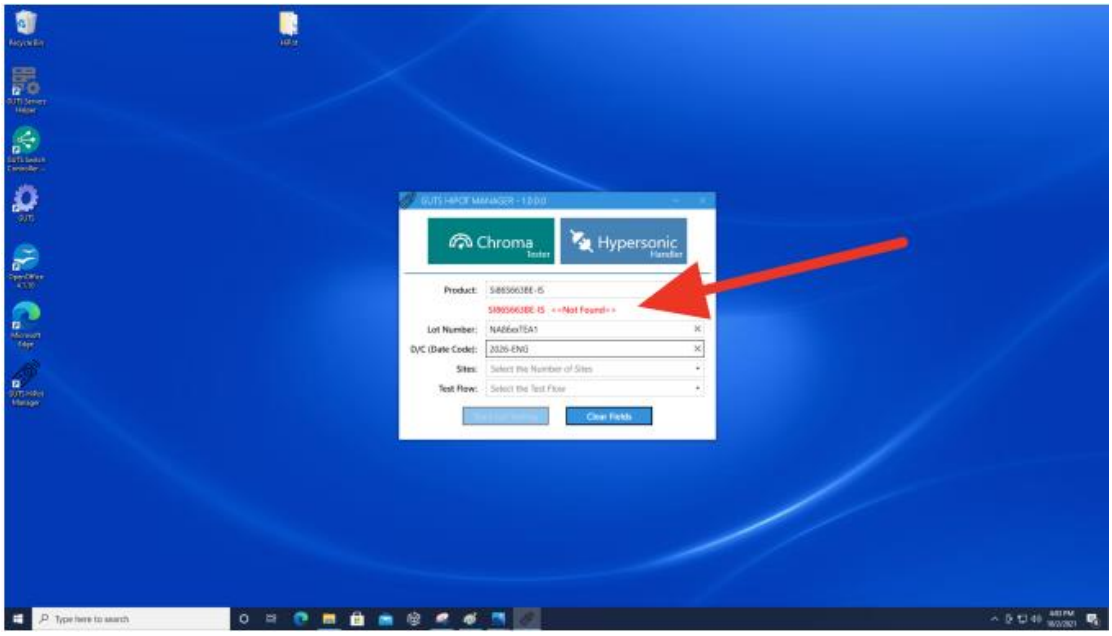


6. Select each field (Product, Lot Number, D/C (Date Code)) one at a time and use the barcode scanner to scan the barcode for the corresponding field on the box of parts:



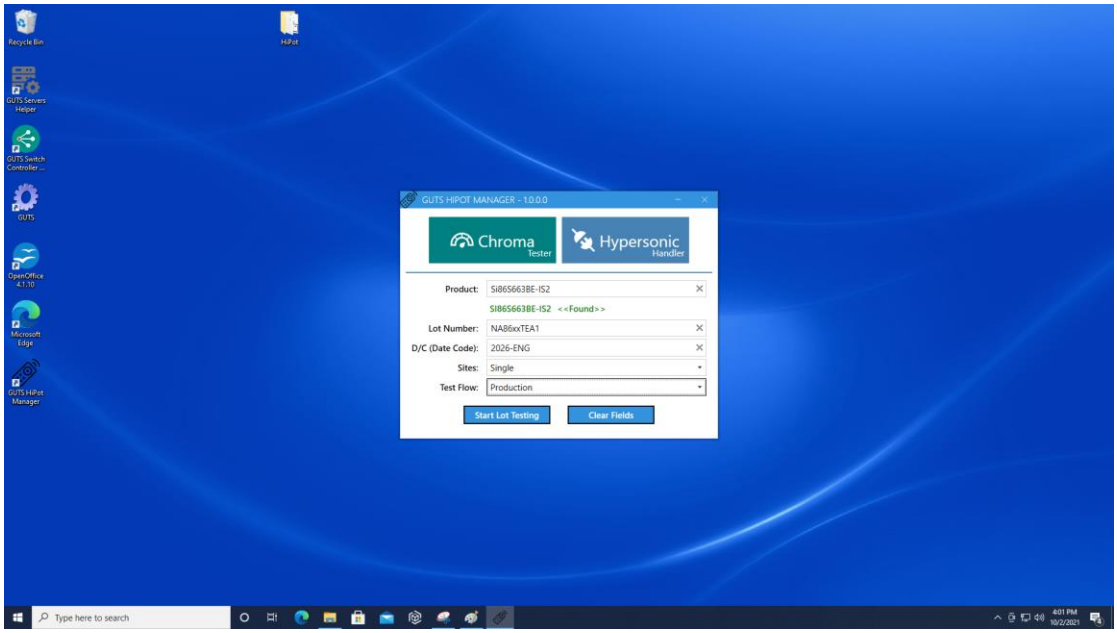
NOTE: These can also be manually entered using a keyboard though it is not recommended

7. If the OPN is incorrect or not found, an error will be displayed:



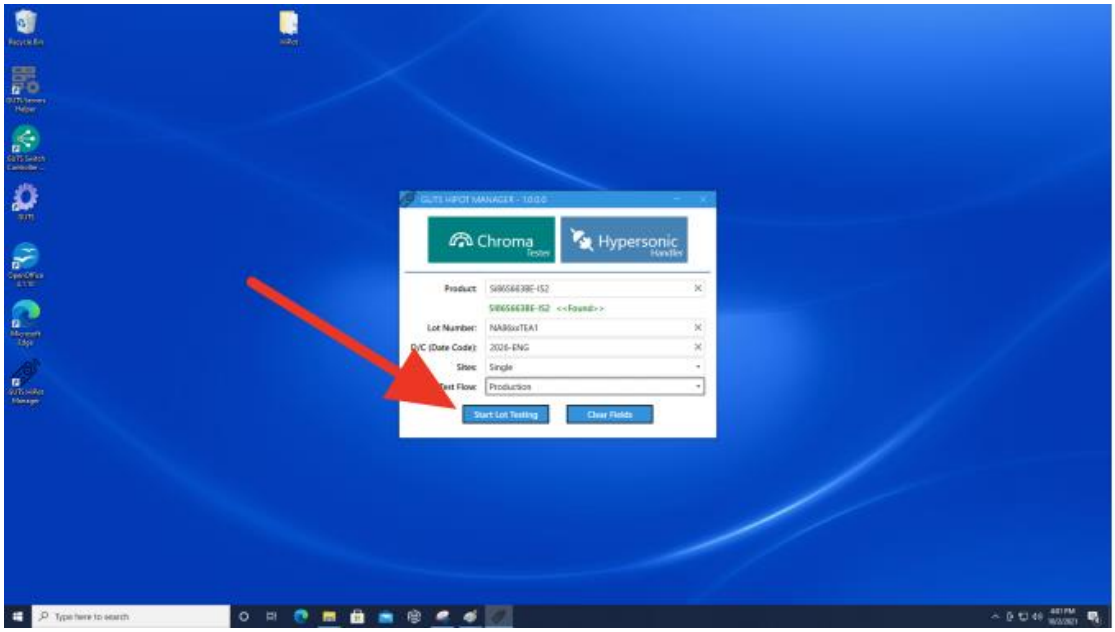
NOTE: If an error is displayed, double check that the OPN is entered correctly, otherwise contact engineering

8. Select the number of Sites and the Test Flow:

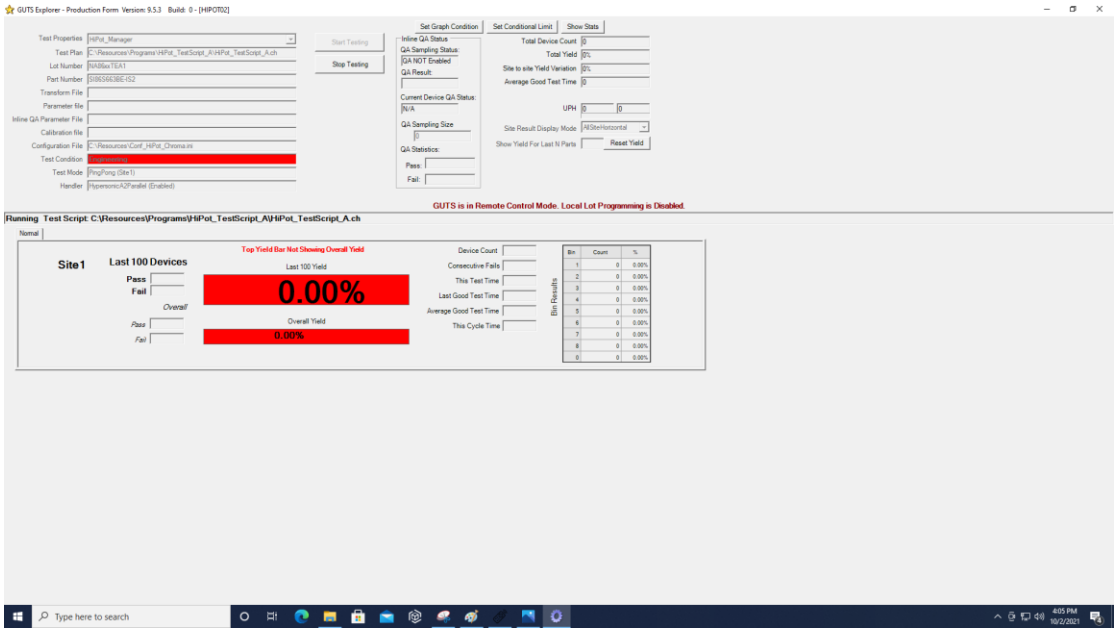


NOTE: For production testing, the Test Flow should always be set to “Production”

9. SKIP FOR NOW – IN PROGRESS - If the “” error appears then the Chroma PD Verification must be completed before you are able to start testing. Change the Test Flow to “” and go to section 5.3 for a walkthrough on how to perform the Chroma PD Verification. Once it is complete (or not required), you must re-do the steps in this section but this time selecting the desired test flow and then proceeding to step 10.
10. Select “Start Lot Testing”:



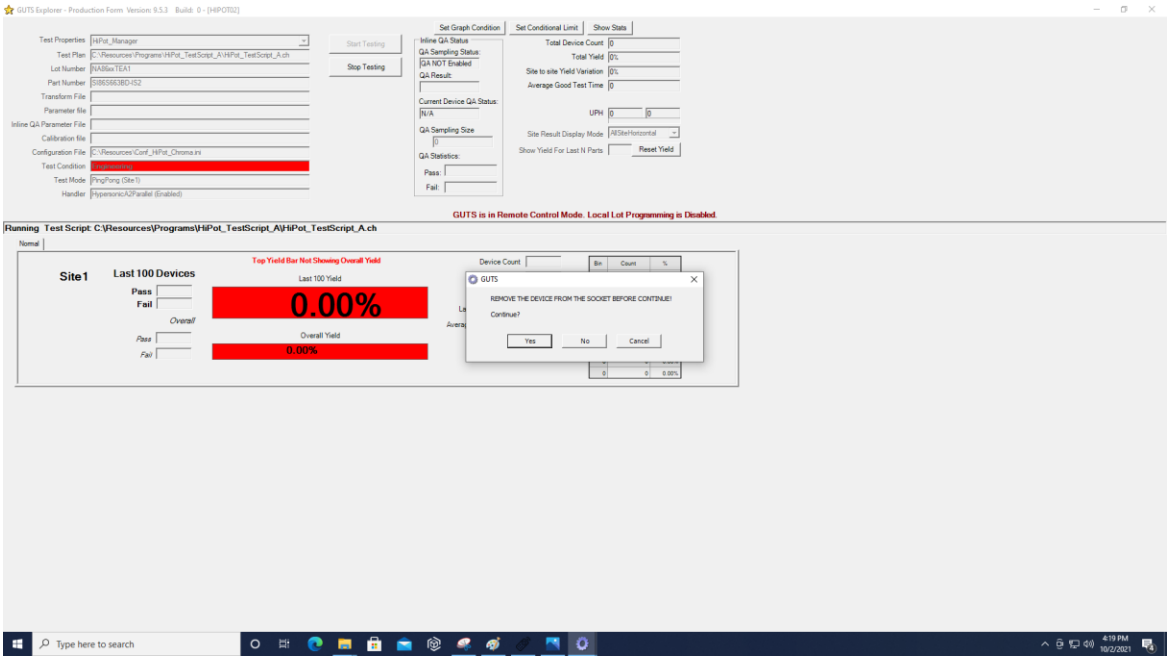
11. The “GUTS Explorer – Production Form” window will open:



At this point, both the Chroma 19501-K and DRVT-1000 have an open socket calibration that needs to be performed at the start of each lot. Section 5.1 outlines the Chroma 19501-K open socket calibration and section 5.2 outlines the DRVT-1000 open socket calibration.

5.1 TESTING PARTS – CHROMA 19501-K OPEN SOCKET CALIBRATION

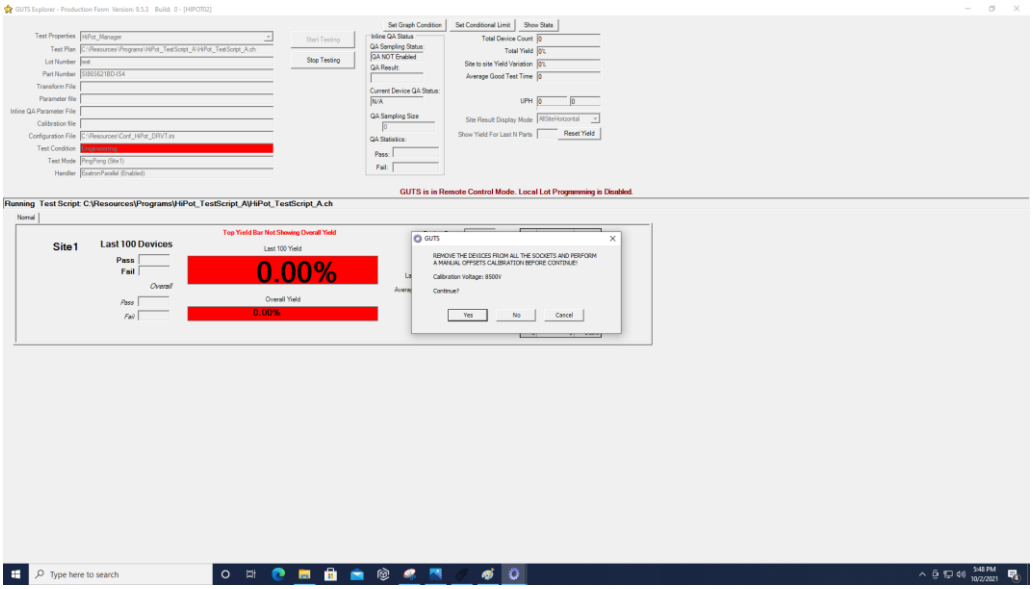
Once the “GUTS Explorer – Production Form” window appears, the program will automatically begin initialization (can take a few seconds). Eventually, the following window should appear:



Make sure that no device is present at the test site and click “Yes” to begin the open socket calibration (NOTE: High Voltage will be present at the test location during this calibration). After ~5 seconds, the open socket calibration should be complete and testing is ready to commence (Section 5.3).

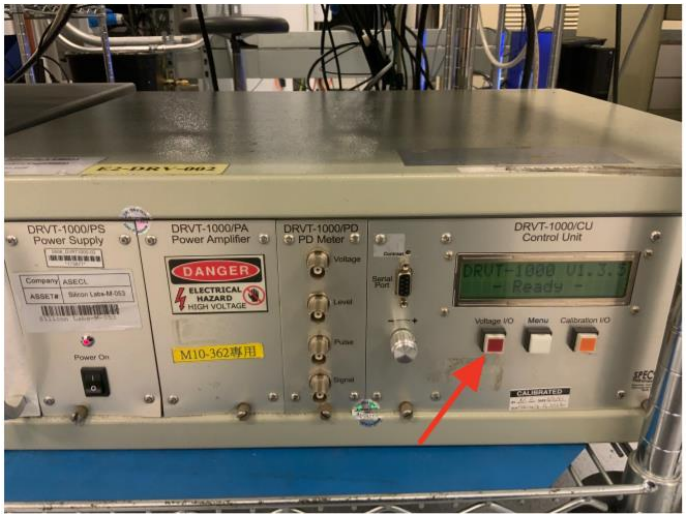
5.2 TESTING PARTS – DRVT-1000 OPEN SOCKET CALIBRATION

Once the “GUTS Explorer – Production Form” window appears, the program will automatically begin initialization (can take a few seconds). Eventually, the following window should appear:

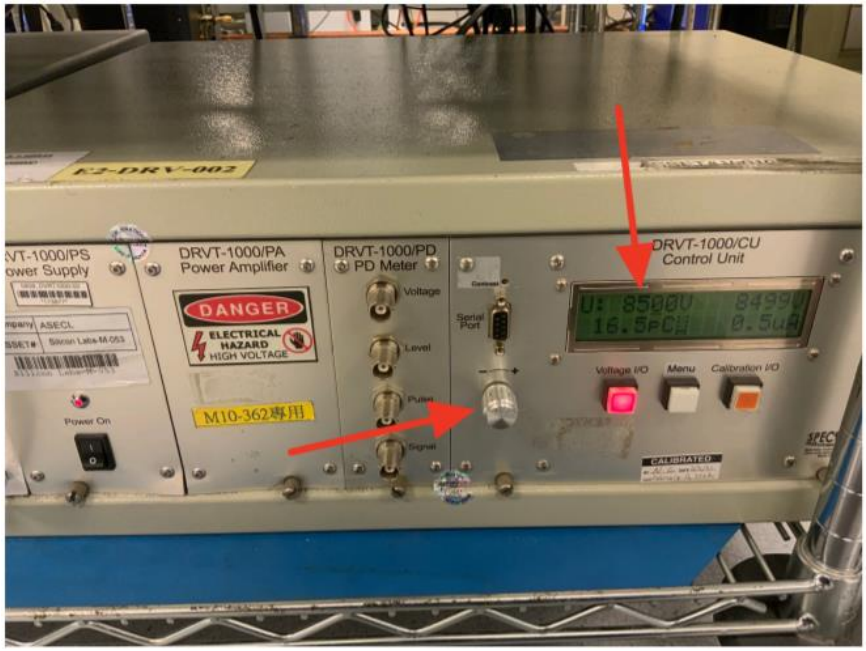


The DRVT-1000 open socket calibration is a manual procedure. Note down the “Calibration Voltage” on the above screen, make sure there is no device present at the test site, and then proceed with the following steps on the DRVT-1000:

1. Press the red “Voltage I/O” button (NOTE: High Voltage will be present at the test location during this calibration):



2. Turn the gray knob to raise the voltage up to the specified “Calibration Voltage” that we noted from before:



3. Once the screen is displaying the correct voltage, press the yellow “Calibration I/O” button to begin the open socket calibration:

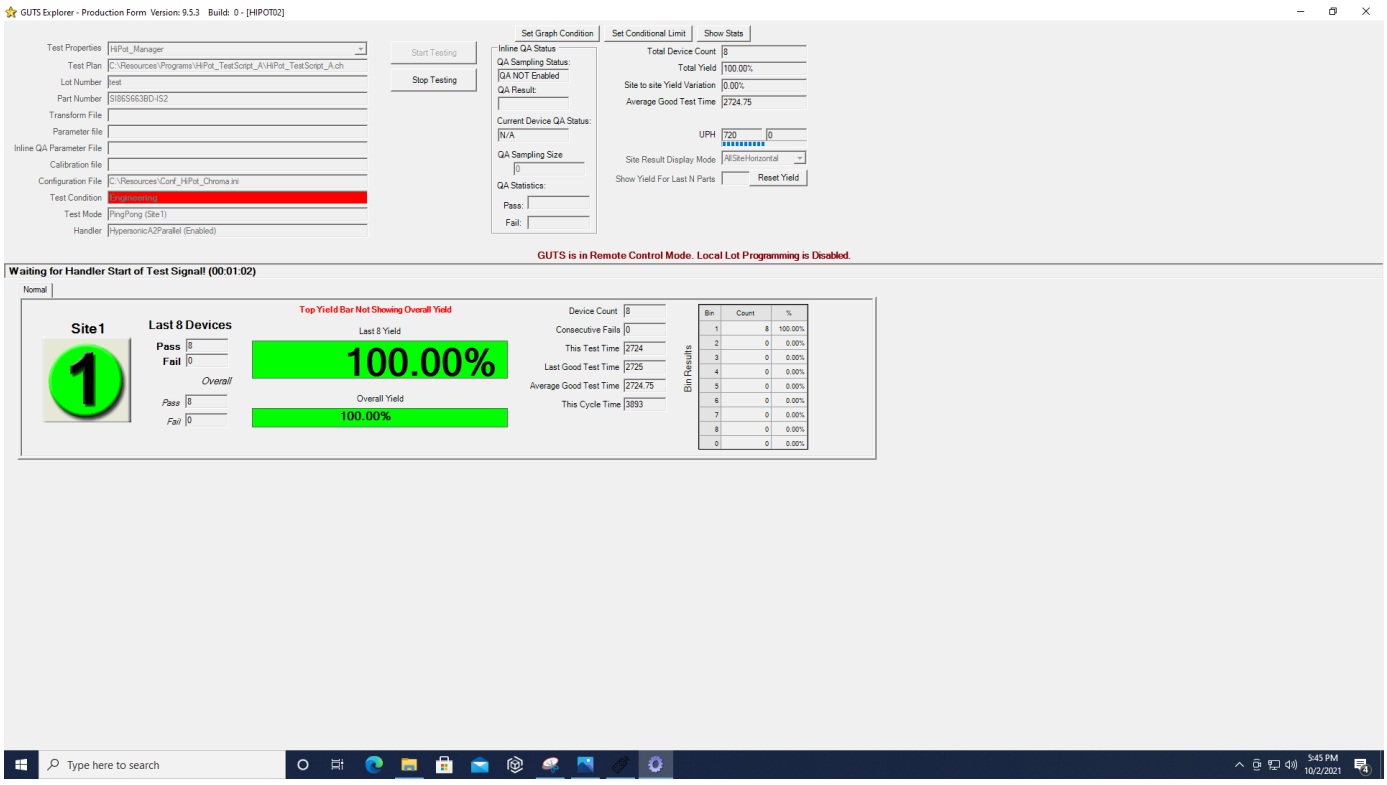
After ~5 seconds, the open socket calibration should be complete and testing is ready to commence (Section 5.3).

5.3 TESTING PARTS – CHROMA 19501-K PD VERIFICATION

In progress

5.4 TESTING PARTS – START TESTING

Now that the system is calibrated, testing can commence. Place the parts on the handler and initiate the handler to begin testing. The “GUTS Explorer – Production Form” window will display the testing results in real time:



5.5 TESTING PARTS – END TESTING AND SAVE DATALOGS

When you are finished testing a lot, click the “Stop Testing” button:

The datalogs should automatically save to the GUTS PC and the “GUTS Explorer – Production Form” window will close.